

Title: Adversarial Telemetry Manipulation in Formula 1: A Cybersecurity Analysis of Race Strategy Prediction

Module: Research Project (CYS6001-20)

Assessment: S2 Research Paper

Github Link: <https://github.com/khadijahkhurum/ResearchProject>

Abstract:

Machine learning is now an important part of the operations of modern Formula 1 and it is used to aid real-time telemetry analysis, race strategy optimisation, and performance prediction. Although such models are widely used, accuracy and efficiency are the main factors considered, and limited attention was paid to their security in conditions of adversaries. This study explores how the racing analytics models can be attacked through adversarial machine learning, particularly the telemetry data manipulation during inference. They provided simulated racing telemetry with small, well-designed perturbations to determine their effect on predictions of machine learning-based strategies. The findings prove that low-magnitude adversarial noise has the power to produce silent prediction failures without activating standard anomaly detection systems. Although the defensive strategies enhanced robustness, it came with trade-offs in latency and false positive rates, which are vital limitations in real time racing set-ups. These results suggest the necessity to consider machine learning models in Formula 1 as security-critical assets and to use adversarial robustness in designing and testing these models.

Keywords:

Adversarial Machine Learning, Formula 1, Telemetry Security, Racing Analytics, Cyber-Physical Systems

Table of Contents

1. Introduction:	5
2. Literature Review:	6
2.1. Adversarial Machine Learning in Safety-Critical Systems:	6
2.2. Adversarial Threats in Autonomous and Automotive Systems:	6
2.3. Adversarial Vulnerabilities in IoT and Edge-Based Telemetry Systems:	7
2.4. Network-Level and Transferable Adversarial Threats:	7
2.5. Defense Mechanisms and Performance Constraints:	8
2.6. Research Gaps and Relevance to Formula 1 Cybersecurity:	8
3. Research Methodology:	9
3.1. Research Design:	9
3.2. Data Source and Collection:	9
3.3. Feature Engineering and Label Definition:	9
3.3.1. Input Features:	10
3.4. Train–Test Split Strategy:	10
3.5. Baseline Model Architecture:	11
3.6. Adversarial Telemetry Tampering Attack:	11
3.7. Threshold-Based Decision Strategy:	12
3.8. Robust Training via Noise Augmentation:	12
3.9. Evaluation Metrics:	12
3.10. Reproducibility and Implementation:	12
4. Results and Analysis:	13
4.1. Baseline Model Performance (Clean Telemetry):	13
4.2. Threshold-Based Baseline Evaluation:	13
4.3. Impact of Adversarial Telemetry Tampering:	13
4.3.1. Performance Under Attack:	14
4.3.2. Robustness Curve Analysis:	14
4.4. Probability Distribution Shift:	15
4.5. Robust Training Evaluation:	15
4.5.1. Robust Model (Clean Test Data):	15
4.5.2. Robust Model Under Adversarial Attack:	15
4.6. Key Findings:	16
4.7. Summary:	16
5. Discussion and Security Implications:	17
5.1. Interpretation of Experimental Findings:	17

5.2. Adversarial Telemetry Tampering as a Realistic Threat:	17
5.3 Confidence Suppression as the Primary Failure Mode:	18
5.4. Effectiveness and Limits of Robust Training:.....	18
5.5 Implications for Formula 1 Analytics Systems:	19
5.6 Broader Cybersecurity Significance:	19
6. Limitations and Threats to Validity:.....	20
7. Conclusion and Future Work:	21
8. Bibliography	22

1. Introduction:

Artificial intelligence and machine learning have transformed the manner in which teams acquire, process, plan and respond to real-time data in Formula 1 racing. Modern telemetry systems deliver thousands of pieces of data per second that then feed the predictive models to guide informed choices of race strategies, performance optimisations and component reliability. However, the use of these automated analytics is also posing novel cybersecurity risks as illustrated by the accumulating body of research indicating that machine learning models particularly those used in a connected or edge environment are susceptible to adversarial attacks. Under these attacks small and calculated perturbations are involved which make models misclassify or break down. (NIST, 2025) (Wang, 2024)

Recent advances in adversarial machine learning have demonstrated that even very high-trained neural networks can be misled by evasion attacks, poisoning attacks or by employing model-extraction techniques. (Kim, 2024) (Chen, 2025) The adversarial ML attacks require particularly greater understanding and reduction of the threats to confidentiality, integrity and availability of Formula 1 telemetry and analytics systems in the high-performance and low-latency domain, such as motorsports, where the decision-making of the races, the integrity and data authenticity in real time must be considered and prevented. This paper directly outlines Formula 1 telemetry-based predicting pit stops as a cybersecurity issue, and how low-magnitude data integrity breaches of real-time decision systems can undermine strategic reliability without activating the typical performance alarms.

2. Literature Review:

2.1. Adversarial Machine Learning in Safety-Critical Systems:

AML Adversarial Machine learning (AML) is the name of a systematic study, as well as a deliberate exploitation of the built-in vulnerabilities of machine learning models by perturbing the input data or the training pipeline. It is also known that AML can include methods that deliberately make a Machine Learning system provide incorrect or misleading results that is often accomplished by making imperceptible alterations to its inputs (NIST, 2025). Such drastic changes though small alter the behaviour of the model drastically.

Evasion attacks are a type of attacks that attacks models in a way that causes them to make wrong classifications when fed with altered inputs. The poisoning attacks jeopardize the integrity of the training data, biases it with malicious samples which decrease the accuracy of the model over time. Model extraction or inversion attacks are aimed at replicating, stealing or estimating sensitive parameters of a machine learning model. (Chen, 2025)

Such attacks have been massively demonstrated in autonomous driving and IoT applications where real time data streams are vital in decision making and strategizing. Unless mitigated, the methods could be used in the context of Formula 1 when it comes to manipulative telemetry or data simulation to make incorrect predictions of race-strategy or misunderstand sensor data.

2.2 Adversarial Threats in Autonomous and Automotive Systems:

Studies have also been especially active in the automotive industry on the topics of Adversarial Machine Learning in which the study of the resilience of LiDAR- based perception models were also carried out and the results showed that physically invisible distortions could radically change the results of object observation. It was further expanded on by (Chen, 2025) through examining the concepts of (deep-learning perception models in real-world driving systems) which revealed the potential of adversarial image distortions to mislead the modules of navigation and decisions. These papers have set a significant precedent - complex, high-speed systems based on sensor fusion are necessarily exposed to AML manipulation. (Khazane & Ridouani, 2024)

Considering the context of Formula 1, where the ML is applied to predictive analytics, but not to direct navigation, even small perturbations may pose critical consequences. An inaccurate signal on the temperature of tires might be an example where false information might be given

to the race engineers, leading them to make wrong strategic decisions. Nonetheless, there is a lack of domain-specific research in the literature that would confirm the way these attacks will occur in the conditions of the actual racing road - the research gap that the present study should fill.

2.3. Adversarial Vulnerabilities in IoT and Edge-Based Telemetry Systems:

The introduction of ML models on resources constrained IoT devices also creates new attack surfaces that were also considered in an extensive taxonomy of adversarial attacks in resource-constrained internet of things environments by (Khazane, 2024) which notes that low processing power tends to disarm defence systems.

These findings have direct implications for Formula 1, where telemetry sensors function as IoT edge devices transmitting high-frequency data to the pit wall and cloud analytics servers. Given the reliance of these systems on lightweight protocols, adversarial data injection could bypass conventional detection mechanisms. The limited bandwidth and strict time constraints in racing make implementing computationally heavy defences particularly difficult, especially adversarial training. (Ibrahim, (2025))

2.4. Network-Level and Transferable Adversarial Threats:

AML is also known to have impacted network-layer defences in addition to the impact on perception and edge devices. This was explored by (Zhang, 2024) that adversarial attacks against intrusion detection system on in-vehicle networks can take place and proved that gradient based manipulation can be used to evade ML-based models of intrusion detection systems.

On the basis of this analysis (Al-Kaf, 2024) has shown that adversarial examples are transferable across models and thus an attacker can also train a surrogate model offline and yet fool the target IDS.

These are essential lessons in Formula 1: Telemetry data communication between the automobile, pit and manufacturer involves the use of encrypted, but semi assistant machine learning algorithms to detect anomalies. The adversarial packet that is made to avoid security systems by masquerading as legitimate data can either generate false alarms or ignore genuine

problems. Literature in the field was also limited particularly in relation to the sustainability of IDS models versus the need to have real time systems. (Yasmine Harbi, 2024,)

2.5. Defense Mechanisms and Performance Constraints:

The primary objectives of defensive AML methods are to improve model robustness or to determine adversarial inputs. Such methods as gradient masking, reducing the ability of the attacker to use gradients to generate adversarial samples, input sanitization, which prunes the input with an anomaly-detection or autoencoder-based system, and adversarial training, which retrain the models on training data with manipulated samples, are common.. (Khazane, 2024)

(Wang, 2024) added that although a more resilient model generally leads to a decrease in the accuracy of the models and a decrease in the computing speed, no definitive defence method can offer complete protection at present. The hybrid control of the anomaly detection system by explainable AI and time series analysis (Li, 2025) has been suggested to provide hybrid control options with Near-Real-Time defence options without significant drops in performance. Hybrid techniques are particularly critical to F1 systems as far as resilience and computational efficiency are concerned.

2.6. Research Gaps and Relevance to Formula 1 Cybersecurity:

Although the fast developments have occurred, most of the adversarial machine learning studies have focused on inert environments, such as autonomous driving or the Internet of Things, as opposed to dynamic, highly dynamic environments such as Formula 1. The key latency-security trade-off that is vital to real-time telemetry systems is not covered by existing frameworks. As well, there is a deficit of literature on the transferability of cross models, or whether attacks trained on simulated race models can be transferred to real race systems. (Wang, 2023)

Additionally, the research shows that it is based on computationally expensive defences that do not suit an embedded racing hardware platform. This work enhances these deficiencies by looking at how the analytics models involved in racing analytics can be vulnerable to adversarial interference and how such interference can be mitigated by the use of lightweight solutions that can be implemented within the time constraints of the F1 racing environment.

3. Research Methodology:

3.1. Research Design:

In this research, the effects of adversarial machine learning evasions on racing analytics models trained on Formula 1 telemetry data are evaluated by means of an experimental study. Specifically, FastF1 Python module, offering Lap timing information, car telemetry information, tire and weather statistics, and session results in the form of structured DataFrame will be used to retrieve real-world data to be used in this study. There are four steps in the methodology:

- Simulation of telemetry data.
- Baseline machine learning models training.
- Adversarial attacks generation.
- Performance analysis and defence evaluation.

As part of adapting them to the constraints of real-time racing information, this plan aligns with prior experimental work on autonomous and Internet of Things systems adversarial machine learning. (Chen, 2025) (Kim, 2024)

3.2. Data Source and Collection:

FastF1 also has a session interface that can be used to download telemetry data: `get_session` (year, event, session). FastF1 exposes timing and telemetry streams through the use of its basic objects. The library offers the analysis of the data through the expansion of the Pandas DataFrames and allows extracting the telemetry data.

3.3. Feature Engineering and Label Definition:

All the data observations made in the collection were identical to the lap made by the racer. A binary type of classification was used to derive the target variable: `PitWithinNextNLaps=1` when the driver has stopped the number of the following N laps. `PitWithinNextNLaps = 0` if not. This final configuration or setting, where $N = 2$, yielded a highly asymmetrical dataset of about 0.54% positive cases which is an indicator of the rarity of pit-stops mid event in the real world.

3.3.1. Input Features:

The feature set was wholly in numeric form and consisted of variables obtained via telemetry, which were:

- Lap number
- Lapping time in seconds.
- Sectors 1, 2, and 3
- Tyre life
- Lap delta relative to the previous lap.
- Such a rapid mean and standard deviation of lap time.
- Tyre compound characteristics that are one-hot encoded.

Non-numerical variables, including the name of the driver, season and the Grand Prix, were eliminated purposely to ensure that information was not leaked as well as to facilitate generalization.

3.4. Train–Test Split Strategy:

Stratified sampling was needed to maintain the extreme class imbalance, and at the same time, keep the valid evaluation, as the dataset was divided equally in both training and test sets to have the same pit-stop events in both.

- Training set: 0.8 (2,068 samples, 11 positives)
- Test set: 20% (518 samples, 3 positives)

This is reflective of the case of operational deployment where models are required to act on very limited positive examples.

3.5. Baseline Model Architecture.

The baseline model was chosen to be a Random Forest classifier because it is robust to feature scaling, interpretable, and it should be used when dealing with tabular telemetry data. The important hyperparameters were:

- 200-400 decision trees
- Balancing of classes by distributing the weight to some extent to counterbalance the class imbalance.
- Simultaneous training in CPU cores.

The telemetry data used to train the baseline model was clean and the performance of the model was measured by F1-score, precision and recall instead of accuracy because of the extremely strong class imbalance.

3.6. Adversarial Telemetry Tampering Attack:

In order to model realistic cyber-physical noise attacks on motorsport telemetry, an additive multiplicative noise attack was applied. The attack disturbs features of numerical inputs in the following manner:

$$x' = x \cdot (1 + \delta), \delta \sim U(-\epsilon, +\epsilon)$$

' ϵ ' being the strength of the attack.

This is also a modelling of the subtle manipulation of sensor streams (e.g. timing jitter, tyre telemetry distortion) as opposed to overt corruption, which is within the constraints of the real world in motorsport systems.

Some of the strengths of the attack considered were:

- $\epsilon = 0.005$ (+-0.5%)
- $\epsilon = 0.01$ (+-1%) $\epsilon = 0.02$ (+-2%)
- $\epsilon = 0.05$ (+-5%)

This threat model is a test-time data integrity attack of a cyber-physical decision system. The attacker is not supposed to be aiming at imposing blatantly wrong predictions, but at undermining confidence calibration with the help of imperceptible telemetry manipulation, thus deteriorating the quality of decisions and avoiding detection.

3.7. Threshold-Based Decision Strategy:

As opposed to use of default class prediction, the model predicted probabilities were strongly kept in a threshold to include more infrequent pit-stop. The probability distributions were inspected and a fixed decision threshold of 0.05 was picked, which permits the model to trade precision against recall in the process of identifying rare strategic events. This is a realistic decision-making within race strategy systems, of which it is more expensive to miss a pit-stop opportunity than to make false positives.

3.8. Robust Training via Noise Augmentation:

In order to estimate defensive mechanisms, a powerful model was trained on noise-augmented data. The training set was increased by union:

- Original clean telemetry samples.
- Noisy version of the same samples generated using $\epsilon = 0.02$.

This essentially doubled the data available to training and subjected the model to adversarial training in the learning process, an approximation of adversarial training with no gradient access.

3.9. Evaluation Metrics:

The following were used to measure model performance:

- F1-score (positive class)
- Precision and recall
- Predicted pit-stop events number.
- Attack-prone predicted probability distributions.

The curve with robustness was plotted, where F1-score was taken as a function of attack strength (ϵ), to determine the levels of failures and measure the degradation in case of telemetry tampering.

3.10. Reproducibility and Implementation:

All experiments were done in Python using scikit-learn and run in a notebook pipeline that was modular. Random seeds were affixed throughout data splitting, noise generation and model training to provide reproducibility.

4. Results and Analysis:

4.1. Baseline Model Performance (Clean Telemetry):

The reference level of performance of the baseline Random Forest model was initially checked on clean telemetry to assess its initial performance level. Accuracy was not regarded as a reliable measure because the ratio of the number of positive samples to the total number of samples is extremely large (0.54%). Rather, F1-score, precision and positive (pit-stop) recall were used. In the rules of default decisions, the baseline model reached:

- Accuracy: 0.994
- F1-score (positive class): 0.000

Although very accurate, the model did not correctly predict any pit-stop events but only majority class. This proves the fact that naive evaluation which is based on accuracy is misleading in strategic racing situations where important events are uncommon.

4.2. Threshold-Based Baseline Evaluation:

In order to capture a more realistic operational decision-making, predicted pit-stop probabilities would be in threshold at 0.05, such that the model would be biased towards recall but not accuracy. The baselines increased with the application of thresholding:

- F1-score: 0.11
- Recall (positive class): 0.33
- Positives predicted: 15 (out of 518 test samples)

This indicates that the model does acquire predictive structure that is useful, but with small confidence margins that are susceptible to probability calibration.

4.3. Impact of Adversarial Telemetry Tampering:

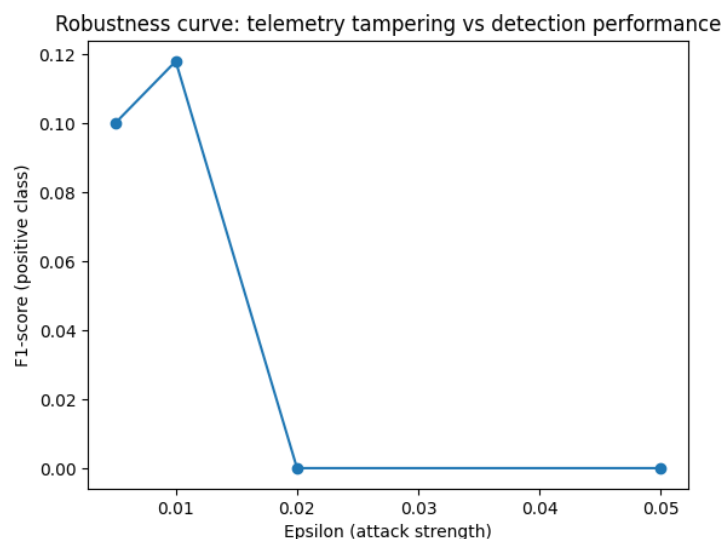
This baseline model was then tested with progressively more telemetric perturbations on it, simulating adversarial manipulation of sensor inputs. The parameter ' ϵ ' was used to control the strength of attacks.

4.3.1. Performance Under Attack:

Attack Strength (ϵ)	F1-score	Predicted Positives
0.005 ($\pm 0.5\%$)	0.100	17
0.010 ($\pm 1.0\%$)	0.118	14
0.020 ($\pm 2.0\%$)	0.000	3
0.050 ($\pm 5.0\%$)	0.000	2

At low perturbation levels ($\epsilon \leq 0.01$), the model retains limited detection capability. However, at $\epsilon = 0.02$, performance collapses entirely, with the F1-score dropping to zero.

4.3.2. Robustness Curve Analysis:



The robustness curve indicates that the performance of detecting the things does not decline slowly, instead it has a sharp non-linear degradation. This implies that there exists a robustness break-point, which is 1-2 percent telemetry tampering. This is a common phase-transition behaviour of brittle decision boundaries in machine learning models, in which small shifts in the inputs are enough to trigger doubt about the less common classes. The probability distribution shift is considered a form of chance variation that occurs even when the underlying mechanism itself remains constant.

4.4. Probability Distribution Shift:

The probability distribution shift is viewed as a type of variation that is caused due to chance, and is the case when the underlying mechanism is held constant. The estimates of predicted pit-stop probabilities are analysed, and it can be seen that adversarial noise does not lead to a high percentage of misclassifications. It rather collimates the predicted probabilities towards zero, moving the true positive samples under the decision threshold. This confirms the attack works through the mechanism of confidence suppression, as opposed to random label flipping a realistic and covert form of failure in the cyber-physical system.

4.5. Robust Training Evaluation:

The noise-augmented telemetry noise-augmented data ($\epsilon = 0.02$) was used to train a robust model, which made the exposure simulate an adversarial exposure.

4.5.1. Robust Model (Clean Test Data):

- **F1-score: 0.10**
- **Predicted positives: 17**

On clean data, the robust model had a similar level of detection capability compared to the baseline, which shows that robustness training did not cause a substantial drop in nominal performance.

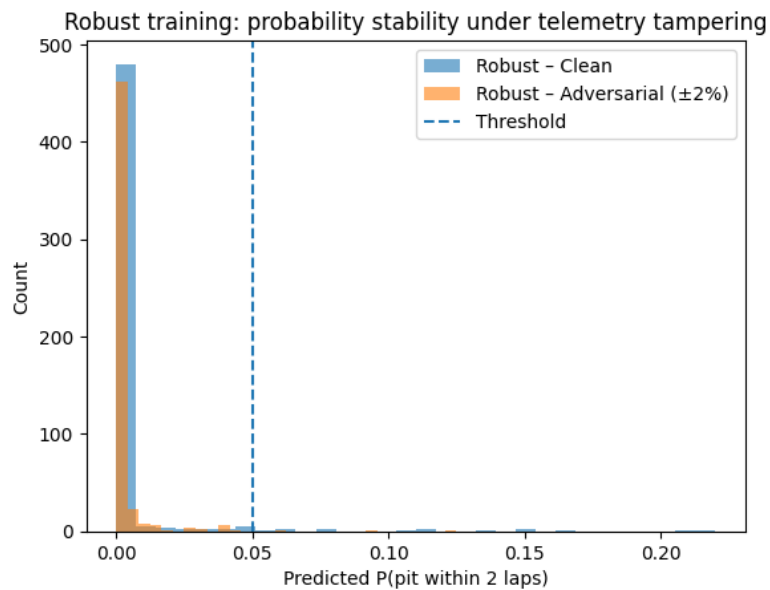
4.5.2. Robust Model Under Adversarial Attack:

At adversarial noise ($\epsilon = 0.02$):

- **F1-score: 0.00**
- **Predicted positives: 3**

Although the robust model has better stability at smaller perturbation magnitude, stronger attacks are still able to overcome it, which indicates that a single noise augmentation is not enough to protect against confidence-based adversarial suppression.

4.6. Key Findings:



The experimental results indicate that:

1. The best measure of strategic racing predictions is inaccuracy because there is too much imbalance in the classes.
2. Baseline model has restricted yet actual predictive property, in case of thresholding.
3. The telemetry tampering of $\pm 2\%$ is also enough to completely neutralise pit-stop detection.
4. Confidence calibration is the main aspect of adversarial noise that impairs feature semantics.
5. Strong training through noise enhancement can enhance resilience only by a small percentage.

4.7. Summary:

On the whole, the findings demonstrate a strong weakness in the race strategy models based on telemetry. Even adversarial perturbations that are not evident can remove the detection ability without any evident anomalies, which presents a serious threat to the use of AI in supporting motorsport decision-making.

5. Discussion and Security Implications:

5.1. Interpretation of Experimental Findings:

The outcomes of the experiment indicate that machine learning models applied to race strategy prediction can have drastic robustness constraints in the face of subtle telemetry perturbation. Though the overall accuracy of the baseline model was large, such measure was deceptive since there is severe imbalance in the number of classes, obscuring the inability of the model to identify infrequent yet strategically important pit-stop events. This proves that decision-support systems functioning in the motorsport settings cannot work with accuracy-focused evaluation. Evaluation using threshold showed the model does learn weak yet informative predictive structure since it is not zero F1-score under clean conditions. Such behaviour is especially worrisome in the security perspective, since it is a silent failure mode where the system is functioning correctly and is actually very precise in general, but becomes a stratagem failure mode when manipulated by an adversary.

Nonetheless, it relies on a limited range of confidence margins, and the model is therefore highly vulnerable to minimal changes in the input features. The robustness curve shows distinct break-even in 1 to 2 percent of telemetry perturbation, and above the threshold, the detection ability falls completely down. This non-linear degradation is an indication that the learned decision-boundary is fragile and prone to confidence-suppression as opposed to the more direct form of misclassification.

5.2. Adversarial Telemetry Tampering as a Realistic Threat:

The telemetry tampering model that was used in this study is a realistic cyber-physical threat scenario. Instead of incorporating conspicuous corruption, the adversarial noise uses small multiplicative perturbation of numerical attributes, which, as a simulation of timing jitter, sensor calibration drift, or malicious interference with telemetry streams. The vulnerability of telemetry-based analytics systems is supported by the fact that even disturbances as minute as $\pm 2\%$ can cancel pit-stop detection.

More importantly, the attack does not cause a significant change in overall accuracy, and so such a manipulation may go unnoticed when using conventional monitoring practices. This is consistent with the tactics in realistic adversarial scenarios, where stealth and plausibility are important factors, instead of disastrous interference. In the Formula 1 setting, any severe telemetry distortion may affect the strategy decisions involving timing of the pit, tyre control, or the undercut implementation, leading to enormous competitive deficit.

5.3 Confidence Suppression as the Primary Failure Mode:

One of the main understandings gained during the probability distribution analysis is that adversarial perturbation is mainly working by suppressing confidence. Instead of switching through predicted labels, the attack shrinks the predicted pit-stop probability values to zero, pushing true positive samples below the operational decision threshold. This mechanism itself is especially effective in highly imbalanced environments, in which the positive-class predictions have already taken up a slender band of the probability space.

This result is important in that it proves that robustness failures can be hidden not as blatant failures to predict, but as subtle changes in model confidence. Consequently, classical validation mechanisms that are based on accuracy or confusion matrices might not detect adversarial degradation, and probability-aware monitoring and evaluation are important.

5.4. Effectiveness and Limits of Robust Training:

The high intensity training through noise augmentation enhanced sensitivity of the model in clean conditions where it predicted more predicted pit-stop events, without a significant detrimental impact on overall performance. This implies that the decision boundaries can be stabilised partially by exposure to perturbed data in training and this enhances resilience to mild noise. Nevertheless, strong training did not help to stop the collapse of performance at the increased attack strengths. The robust model had the same failure mode as the baseline under adversarial perturbation of $\epsilon \geq 0.02$, which implies that basic noise-based defences are not sufficient to defend against confidence-based attacks. This supports the results of defensive strategies in adversarial machine learning literature that defensive strategies need to specifically aim at probability calibration and stabilizing decision boundaries instead of data augmentation.

5.5 Implications for Formula 1 Analytics Systems:

The findings of this paper directly apply to the application of machine learning models in Formula 1 race strategy settings. High-value attack surface Telemetry streams are prone to exploitation, and narrow-confidence-margins models are especially susceptible to exploitation. Since the timing of pit-stop is a strategic issue, even a small and brief manipulation of telemetry may affect the race results with no visible system alarms.

Moreover, it can be inferred that the break-point of robustness needs to be considered not only in the nominal setting, but in a range of plausible perturbations. Non-adversarial system design experiments exaggerate the reliability and resilience of the system in an adversarial or hostile environment.

5.6 Broader Cybersecurity Significance:

In addition to motorsport, the results are relevant to other time-sensitive cyber-physical systems, such as intelligent transportation, industrial control, and IoT-based decision support systems. Rare-event prediction can be useful in such systems, whereas aggregate accuracy can be less important, and confidence suppression attacks can be more dangerous than conventional attacks based on misclassification.

This paper illustrates why it is crucial to consider the robustness of a system based on domain-relevant metrics and threat models and why defensive strategies should consider adversarial manipulation of sensor-based data streams in an explicit way.

6. Limitations and Threats to Validity:

This research has a number of limitations which impact on the overall generalisability of the research. To begin with, the data is highly unbalanced, with the 14 positives pit-stops events, which implies that minority-class measures cannot be as stable, and the variation of the sampling is highly sensitive. This will be a limitation to statistical power, although this is a realistic representation of the rarity of pit stops in the real world. Second, the analysis of a restricted race situation is involved and thus fails to reflect strategic variability that comes about due to variation of weather, deploying safety-cars, or team-specific operation behaviour.

Third, random multiplicative perturbations of numbers to numerical features are used to implement adversarial threat model. Even though this constitutes a realistic type of telemetry tampering, like calibration drift or gentle signal manipulation, it does not model the adaptive attackers who directly optimise the perturbation to maximise the prediction error. Fourth, the tabular Random Forest uses a tabular model, which excludes temporal dependencies that can be used by sequential models. Lastly, the fixed probability threshold is an operation trade-off but cannot necessarily be generalised across deployment situations. Irrespective of these shortcomings, the study is still valid in estimating the robustness break-points with realistic, low-amplitude telemetry manipulation.

7. Conclusion and Future Work:

The study tested the strength of the Formula 1 race strategy prediction in presence of adverse telemetry manipulation. Based on the FastF1 derived race data, the pit-stop prediction was developed as a rare-event classification problem. Measuring baseline performance showed that accuracy is misleading in case of extreme imbalance of classes whereas probability thresholding showed that there is limited but significant predictive power. Under adversarial perturbation, the performance was non-linearly decreasing: detection was still partially stable at $\varepsilon \leq 0.01$ but completely failed at $\varepsilon \geq 0.02$, which demonstrated a distinct break-point of robustness at around +2% telemetry tampering. Noise-enhanced robust training increased sensitivity in clean conditions and was not able to resist collapse in the case of more severe attacks, which validates confidence suppression as the most frequent failure mechanism. These results support the argument that machine learning components of cyber-physical systems should be seen as security-critical assets, and not necessarily as predictive tools.

Future research would aim to increase the assessment of various seasons and circuits, introducing contextual factors, including safety-car periods and tyre plans. More advanced models of adversarial attacks, such as optimisation based and feature constrained ones, would be more realistic of actual threat actors. On the defensive side, probability calibration, cost-sensitive learning and sequential modelling over lap histories are promising directions. There are also operational safeguards that can be considered to reduce strategic risk, including telemetry integrity checking and anomalies detection.

8. Bibliography

1. I. Al-Kaf, A. R. F. M. I. a. O. A., 2024. Adversarial challenges in network intrusion detection systems: research insights and future prospects. *arXiv preprint arXiv:2409.18736*..
2. Chen, C. W. Y. M. N. Z. X. a. Z. X., 2025. *Revisiting Adversarial Perception Attacks and Defense Methods on Autonomous Driving Systems*. [Online]
Available at: <https://arxiv.org/abs/2505.11532>
[Accessed 20 October 2025].
3. Ibrahim, A. D. M., (2025). Deep learning adversarial attacks and defenses in autonomous vehicles: a systematic literature review from a safety perspective. *Artificial Intelligence Review*, Volume 58, pp. 28-35.
4. Khazane, H. & Ridouani, M., 2024. A Holistic Review of Machine Learning Adversarial Attacks in IoT Networks. *Adversarial Attacks in IoT Networks*., Volume 16, pp. 32-28.
5. Khazane, K. A. M. U. S. a. A. A., 2024. A Holistic Review of Machine Learning Adversarial Attacks in IoT Networks. *Future Internet*, 16(1), p. 32.
6. Kim, J. a. K. A., 2024. *A Survey on Adversarial Robustness of LiDAR-based Machine Learning Perception in Autonomous Vehicles*. [Online]
Available at: <https://arxiv.org/abs/2411.13778>
[Accessed 20 October 2025].
7. Li, Y. Z. H. C. J. a. L. X., 2025. *Robust and explainable detection of time-series anomalies using multiclass pseudo-anomaly augmentation*. [Online]
Available at: <https://arxiv.org/abs/2501.09033>
8. NIST, N. I. o. S. a. T., 2025. *Adversarial Machine Learning: A Taxonomy and Terminology of Attacks and Mitigations*, NIST AI 100-2e2025.. [Online]
Available at: <https://nvlpubs.nist.gov/nistpubs/ai/NIST.AI.100-2e2025.pdf>
[Accessed 20 October 2025]].
9. Wang, G. Z. C. W. Y. C. B. G. H. a. Y. Q., 2023. *Beyond Boundaries: A Comprehensive Survey of Transferable Attacks on AI Systems*. [Online]
Available at: <https://arxiv.org/abs/2311.11796>
[Accessed 20 October 2025].
10. Wang, X. L. Y. Z. Z. a. H. F., 2024. Deep Learning Model Security: Threats and Defenses. *arXiv* , Volume 2412, pp. 69-89.
11. Yasmine Harbi, K. M. C. G. Z. A. a. S. H., 2024,. Roadmap of Adversarial Machine Learning in Internet of Things-Enabled Security Systems.. *Sensors*, Volume 24, pp. 51- 50..
12. Zhang, Y. S. L. W. S. a. C. Z., 2024. Adversarial attacks on intrusion detection systems in in-vehicle networks of connected and autonomous vehicles. *Sensors*, 24(3848), p. 12.